

SOHCAHTOA

Foundation Tier

Right-angled trigonometry and applications

Name

Date

10 questions

28 marks

2017-2025

Instructions

- Answer each question in the space provided.
- Show all your working and follow the rounding instructions.
- Diagrams are not accurately drawn unless stated otherwise.
- A calculator may be used for all questions in this booklet.

Using this booklet

Questions are arranged by exam series. Each page gives its original paper reference and question number. The question numbers within the extracted artwork are the original exam numbers.

This is an independently compiled practice resource, not an official Pearson examination paper. The selected questions run through June 2025; November 2025 is not included.

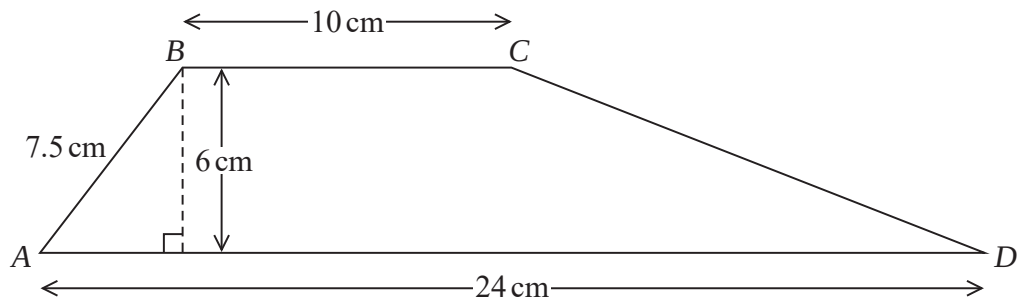
Question index

Foundation Tier | 10 questions | 28 marks

Practice	Exam series	Paper	Original question	Marks	Page
01	November 2017	1MA1/2F	Q22	5	3
02	June 2018	1MA1/3F	Q23	3	4
03	November 2018	1MA1/3F	Q25	4	5
04	June 2019	1MA1/2F	Q24	2	6
05	November 2020	1MA1/2F	Q25	2	7
06	November 2021	1MA1/2F	Q26	4	8
07	November 2022	1MA1/2F	Q22	2	9
08	November 2023	1MA1/2F	Q29	2	10
09	November 2024	1MA1/3F	Q24	2	11
10	June 2025	1MA1/2F	Q26	2	12

The original exam numbering is retained in each question.

22 $ABCD$ is a trapezium.

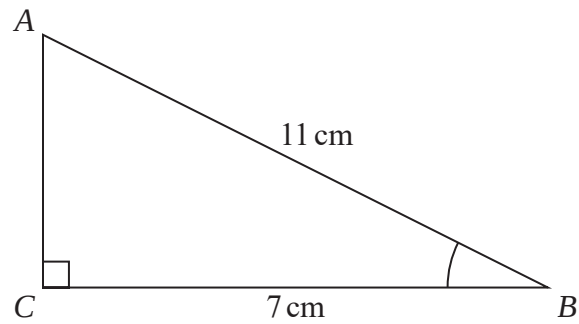


Work out the size of angle CDA .

Give your answer correct to 1 decimal place.

(Total for Question 22 is 5 marks)

23 ABC is a right-angled triangle.



- (a) Work out the size of angle ABC .
Give your answer correct to 1 decimal place.

.....
(2)

The length of the side AB is reduced by 1 cm.

The length of the side BC is still 7 cm.

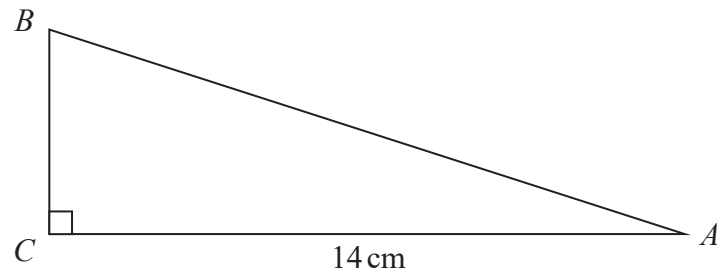
Angle ACB is still 90°

- (b) Will the value of $\cos ABC$ increase or decrease?
You must give a reason for your answer.

.....
(1)

(Total for Question 23 is 3 marks)

25 ABC is a right-angled triangle.



$AC = 14\text{ cm}$.

Angle $C = 90^\circ$

size of angle B : size of angle $A = 3 : 2$

Work out the length of AB .

Give your answer correct to 3 significant figures.

.....cm

(Total for Question 25 is 4 marks)

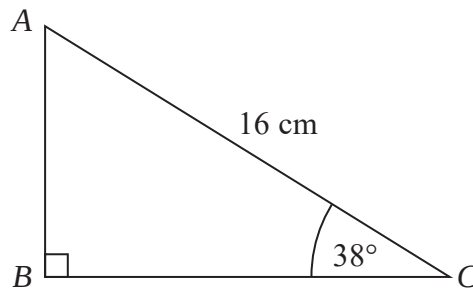
Additional working space

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24 ABC is a right-angled triangle.

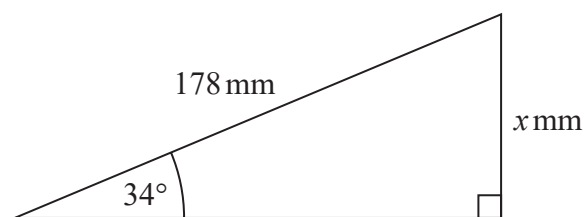


Calculate the length of AB .
Give your answer correct to 2 decimal places.

.....cm

(Total for Question 24 is 2 marks)

Additional working space

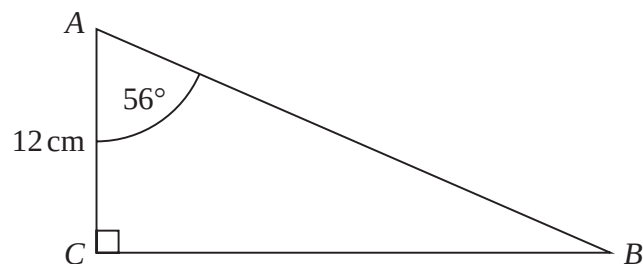


Work out the value of x .
Give your answer correct to 1 decimal place.

(Total for Question 25 is 2 marks)

Additional working space

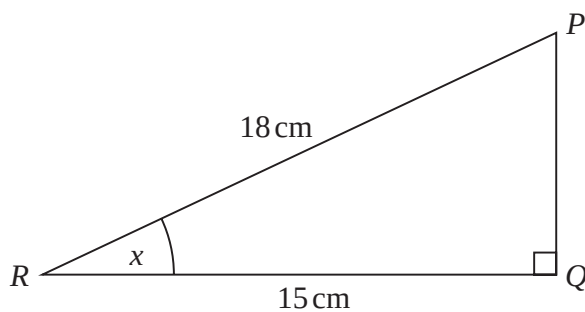
26 ABC is a right-angled triangle.



- (a) Work out the length of BC .
Give your answer correct to 1 decimal place.

..... cm
(2)

PQR is a right-angled triangle.

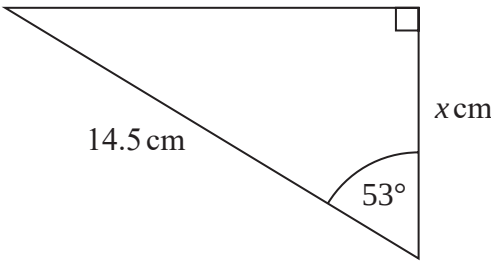


- (b) Work out the size of the angle marked x .
Give your answer correct to 1 decimal place.

.....
(2)

(Total for Question 26 is 4 marks)

22



Work out the value of x .
Give your answer correct to 3 significant figures.

$x =$

(Total for Question 22 is 2 marks)

Additional working space

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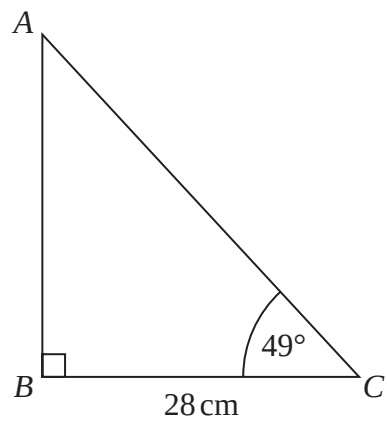
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29 ABC is a right-angled triangle.



Calculate the length of AB .
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 29 is 2 marks)

Additional working space

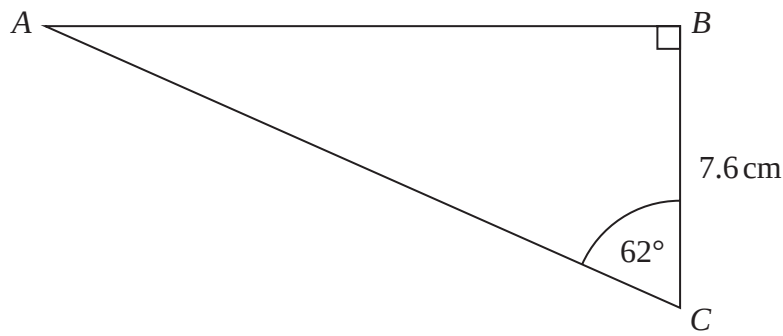
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24 ABC is a right-angled triangle.



Calculate the length of AB .
Give your answer correct to 1 decimal place.

..... cm

(Total for Question 24 is 2 marks)

Additional working space

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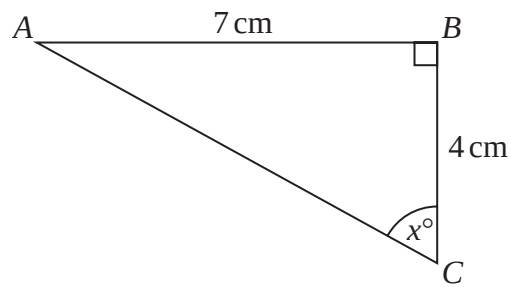
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26 ABC is a right-angled triangle.



Calculate the value of x .
Give your answer correct to 1 decimal place.

$\chi =$

(Total for Question 26 is 2 marks)

Additional working space